Quantum Physics 1

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Abstract

This course was taught by Dr. Liam Fitzpatrick at Boston University during the Spring 2025 semester. We used the text Quantum Mechanics: A Paradigms Approach by David McIntyre. We also use the course website physics.bu.edu/~fitzpatr.

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Chapter 1

Introduction

Lecture 1: Stern-Gerlach Experiments

Tue 21 Jan 2025 14:00

The Stern-Gerlach experiments were done in the 20s, while quantum physics was still being developed. We are given a set of silver atoms, which we will consider for this experiment to be an extremely tiny bar magnet. It has a dipole magnetic moment (vector) $\boldsymbol{\mu}$. The experiment takes a large magnet (image in notes, ill draw later) with a nonconstant magnetic field between north and south. The silver atom is then shot through the $\mathbf{B}$ field, and we look at where it comes out.

Classical mechanics says the magnetic moment $\boldsymbol{\mu}$ passing through the field $\mathbf{B}$ should deflect it as follows. Energy is given by

$$E = -\boldsymbol{\mu} \cdot \mathbf{B}.$$

We then get the force

$$\mathbf{F} = -\nabla E.$$

For example, if $\mathbf{B} = B_z(z)\hat{\mathbf{z}}$ is slightly varying along the z direction, then

$$\mathbf{F} = -\partial_z B_z \mu_z.$$

Note here that ∂_z is the partial wrt z. New notation!!!

The force then depends only on the component of the magnetic moment in the z-direction. $F \propto \boldsymbol{\mu} \cdot \hat{\mathbf{z}} = |\boldsymbol{\mu}| \cos\theta$, where θ is the angle formed with the z-axis. So

$$F_z = |\boldsymbol{\mu}| \frac{\partial B_z}{\partial z} \cos\theta.$$

Since the angle of the moments is distributed randomly, we would expect the distribution of resultant locations to be constant, since each magnetic moment corresponds to a new location, and since the angles should be distributed evenly in θ, they should be distributed evenly through the locations. However, even if they weren't *purely* random, we would get *some distribution.* It would be weird if we didn't see a distribution. So of course we don't see a distribution, because physics.

When we actually do the experiment, they only ever come out at two points, a top point and a bottom point. Since we know what $B_z(z)$ is, and since we know the mass of a silver atom, we can find the force and thus find the exact magnetic moment. We find that $\mu_z = \pm |\boldsymbol{\mu}|$ always. Even though the angles were randomly distributed, it always shows up as two possibilities.

We say it is quantized. In classical mechanics, we have a continuous distribution, but in quantum mechanics, the possibilities are all quantized into two possibilities. In quantum physics, we will talk a lot about the state of a system, and $|\Psi\rangle$ will denote the state of some system. The symbol $|\cdot\rangle$ is called a "ket".

Classically, the state is the direction of the magnetic moment, since that determines all the information we need. However, in quantum mechanics, we cannot describe the state of a system using classical physics, which is why we use kets to denote it. They are a completely new thing that doesn't show up in classical mechanics, and we will describe precisely what a ket is as we continue.

Since there are only two possibilities in the experiment, there must be only two states. We can find that those two states are

- $|+\rangle$ is the state where $\mu_z = +|\boldsymbol{\mu}|$;
- $|-\rangle$ is the state where $\mu_z = -|\boldsymbol{\mu}|$;

where again $|\boldsymbol{\mu}|$ is the moment of a silver atom.

Another thing to note is that the distribution is even throughout the possible states; 50% of the time we see $|+\rangle$, and the rest of the time we measure a state of $|-\rangle$.

Experiment two takes all the atoms that have already been deflected, and passes them through *another* magnetic field. (draw later) All the up ones keep going up, and all the down ones keep going down. Basically we didn't change anything so it stays the same. Makes sense. Ok technically we only re-measure $|+\rangle$ states.

Experiment 3 starts off the same way as experiment 1, and then proceeds *similarly* to experiment 2, but instead of measuring them in the $\hat{\mathbf{z}}$ axis again, we change the magnetic field to point in the $\hat{\mathbf{x}}$ direction. The z direction isn't special or anything, so we can do this. Classically, we would note that $\mu_z = |\boldsymbol{\mu}|$, so we would assume that $\mu_x = 0$, since the entire moment is used up in the z-component. BUT of course we get a weird answer. We once again see exactly two spots hit; exactly 50% of the time they end up in the $+x$-direction, and 50% of the time they go to the $-x$-direction. So the conclusion is the moment has multiple distinct components; orthogonal directions correspond to distinct components. We call these states $|+\rangle_x$ and $|-\rangle_x$, respectively.

Experiment 3 continues by measuring the atoms once again. So we measured them in z and found them to be distributed 50/50 in two spots. We took the $|+\rangle$ state and measured them in the x-direction and found another 50/50 distribution. However, we already measured them as $|+\rangle$ in the z-direction, so we expect them to still be in this state. This is like if we took a black sock, sorted it based on whether it was striped or not, then checked to see if it was still black. However, we *DON'T SEE THIS*. We see instead another 50/50 distribution. The particles are no longer still $|+\rangle$, but instead it's back to a completely random distribution.

So what does this mean? Measuring in z repeatedly keeps it in the same state, but when we measured it in the x component, it changed. Later, we will explain how we know this is actually what happens, and how we know this isn't the experimental machine just being shit. So measuring it in one component changed the other component. Additionally, if we only change the direction a little, rather than orthogonally, we still measure a new distribution, it's just not 50/50. So it is the act of measurement itself that changes the state of the particle. In classical physics, measuring something doesn't change it. Welcome to quantum physics, where the state $|\Psi\rangle$ cannot have a well-defined magnetic moment in two directions at the same time.

One last point, which we will discuss later. The magnetic moment $\boldsymbol{\mu}$ is related to the spin $\mathbf{S}$ by the relation

$$\boldsymbol{\mu} = \frac{ge}{2m_e}\mathbf{S}.$$

The symbol g is some constant. The spin $\mathbf{S}$ is the angular momentum of the electron when it's sitting still. Moment pointing up correspnds to

$$S_z = \frac{\hbar}{2}.$$

Definition 1.1: Kronecker Delta

The Kronecker Delta is denoted δ_{ij}, and is defined as

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}.$$

We can interpret δ_{ij} as the identity matrix $\mathbb{1}$, since it has 1s only when the indices $i = j$. Let $\mathbf{a}$ be a vector. Then $\delta_{ij} a_j = a_i$ (in Einstein notation).

Lecture 2: Linear Algebra

Thu 23 Jan 2025 14:00

The state of a system encodes all information about a system. We denote this by $|\psi\rangle$ in quantum physics. Also note that when an index is shared over two elements:

$$M_{ik} N_{kr},$$

this implies this expression really indicates a summation over the shared index:

$$M_{ik} N_{kr} \coloneqq \sum_k M_{ik} N_{kr}.$$

A ket $|\psi\rangle$ represents the state of some system, but it is also a vector in a $\mathbb{C}$-vector space. This ket notation is useful since we may deal with infinite-dimensional spaces, and it also plays nicely with inner products. We always assume a vector space in quantum mechanics is a space in $\mathbb{C}$-**Vect**.

The question is, what really is $|\psi\rangle$, and how do we get information out of it? In order to do this, we need to equip our space with an inner product, making it a Hilbert space. Note that all n-dimensional $\mathbb{C}$-Hilbert spaces can be given as the matrix space $M_{n\times 1}(\mathbb{C})$. We then define $|\psi\rangle^\dagger \coloneqq \pi^{-1}(\pi(|\psi\rangle)^\dagger)$, for $\pi : \mathcal{H} \to M_{n\times 1}(\mathbb{C})$ the isomorphism. More simply, given a vector

$$\begin{pmatrix} x \\ y \end{pmatrix},$$

define

$$\begin{pmatrix} x \\ y \end{pmatrix}^\dagger = \begin{pmatrix} x^* & y^* \end{pmatrix}.$$

Equivalently,

$$(z_1 |v_1\rangle + z_2 |v_2\rangle)^\dagger = z_1^* \langle v_1| + z_2^* \langle v_2|.$$

The Hermitian conjugate $|\psi\rangle^\dagger$ is denoted as a 'bra', $\langle\psi|$. Note that given vectors $|\phi\rangle, |\psi\rangle$, we can define an inner product

$$|\phi\rangle \cdot |\psi\rangle \coloneqq \langle \phi \,|\, \psi \rangle.$$

In a more general vector space, we still denote the inner product as $\langle \phi \,|\, \psi \rangle$.

Definition 1.2: Inner Product

An inner product is a map $\langle \cdot \,|\, \cdot \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$ such that

- $\langle v_1 \,|\, v_2 \rangle = \langle v_2 \,|\, v_1 \rangle^*$;
- $\langle v \,|\, v \rangle \geq 0$;
- $\langle v \,|\, v \rangle = 0$ iff $|v\rangle = 0$;
- $(a\langle v_1| + b\langle v_2|) \cdot (c|w_1\rangle + d|w_2\rangle) = ac\langle v_1 \,|\, w_1\rangle + ad\langle v_1 \,|\, w_2\rangle + bc\langle v_2 \,|\, w_1\rangle + bd\langle v_2 \,|\, w_2\rangle$.

Our way of pulling information out of a ket will be by taking its inner product with a state we already understand. For example, in the Stern-Gerlach experiments, we knew the state $|+\rangle$ indicated $\boldsymbol{\mu}$ pointing up, and $|-\rangle$ indicated pointing down. Assume $|+\rangle, |-\rangle, |\psi\rangle$ are all normalized with norm 1; $\langle \psi \,|\, \psi \rangle = 1$. For some reason, $|+\rangle, |-\rangle$ form a basis. We can then find the inner product of ψ with any vector using this fact. The Born Rule tells us what information this actually gives us; we will state it here, specialized to the Stern-Gerlach experiments.

Proposition 1.1

The probability of finding the state $|\psi\rangle = |+\rangle$ is

$$P_+ = |\langle + \,|\, \psi \rangle|^2 .$$

Similarly, the probability of finding the state $|\psi\rangle = |-\rangle$ is

$$P_- = |\langle - \,|\, \psi \rangle|^2 .$$

Moreover, when we measure $|\psi\rangle$ in one of these states, it turns into that state.

Since the probabilities have to sum to 1, we assume ψ has a unit norm. This follows since $\langle + \,|\, - \rangle = 0$, so the probability of finding it at both $|+\rangle$ and $|-\rangle$ is 0.

Since $|+\rangle, |-\rangle$ forms a complete basis, we can write

$$|\psi\rangle = \langle + \,|\, \psi \rangle |+\rangle + \langle - \,|\, \psi \rangle |-\rangle.$$

Then

$$\| |\psi\rangle \| = |\langle + \,|\, \psi \rangle|^2 + |\langle 0 \,|\, \psi \rangle|^2 .$$

Lecture 3: owo

Tue 28 Jan 2025 14:01

The kets $|+\rangle, |-\rangle$ form a complete basis because they are the only measurable states, and the vector space is the space of all states.

The born rule says

$$P_\pm = |\langle \pm \,|\, \psi \rangle|^2 .$$

That is, the probability of measuring a given state is the same as the norm squared of the inner product with the state vector ψ with the observable state $|\pm\rangle$.

We can also change the direction in which we measure from the $\hat{z}$ direction to the $\hat{x}$ direction. We write $|\pm\rangle_x$. We can assume WLOG that these states are normalized; that is, $\langle + \,|\, + \rangle_x = 1$. Moreover, $\langle + \,|\, - \rangle_x = 0$, because we cannot measure one state and then have it change in repeated experiments with no change. This means state vectors are orthogonal. Everything about the x-states is the same as the z-states.

It's difficult to describe how to desccibe a well-defined z and x direction at the same time; how can we show what direction $|\psi\rangle$ points in given only 2 states?

Note that if you measure $|+\rangle$ in the z-direction, then you have a 50% chance of measuring in $|+\rangle_x$ or $|-\rangle_x$, so we have that

$$P_+^{(x)} = P_-^{(x)} = \frac{1}{2}.$$

It follows that

$$|\psi\rangle_x = C_+^{(x)} |+\rangle_x + C_-^{(x)} |-\rangle.$$

In this case, we have

$$P_\pm^{(x)} = \frac{1}{2} = |\langle +_x \,|\, + \rangle|^2 = |\langle -_x \,|\, + \rangle|^2 ,$$

because measuring $|+\rangle$ gives a $1/2$ chance of measuring either $+_x$ or $-_x$. Since $C_\pm^{(x)} \in \mathbb{C}$, we have that

$$\left|C_\pm^{(x)}\right| = \frac{1}{\sqrt{2}}.$$

This is because

$$C_\pm^{(x)} = \langle \pm_x \,|\, + \rangle.$$

This follows from orthogonality. Since these coefficients are complex-valued, this only tells us the probability, but we cannot discern the phase of these states. The phase of a complex number z is the θ such that $z = re^{i\theta}$. We only know the magntidude, r. This is as far as we can go experimentally, and so we can arbitrarily choose values for $C_\pm^{(x)}$ without loss of generality, so we define

$$|+\rangle = \frac{1}{\sqrt{2}}|+\rangle_x + \frac{1}{\sqrt{2}}|-\rangle_x.$$

We can use the same logic to find that the magntitude of the coefficients

$$|-\rangle = C_+^{(x)}|+\rangle + C_-^{(x)}|-\rangle$$

are also $1/\sqrt{2}$. However, we also know that since $\langle + \,|\, - \rangle$ are orthogonal, we have that

$$\langle - \,|\, + \rangle = \frac{C_+^{(x)}}{\sqrt{2}}\langle +_x \,|\, +_x \rangle + \frac{C_-^{(x)}}{\sqrt{2}}\langle -_x \,|\, -_x \rangle = 0.$$

We choose the convention that

$$C_+^{(x)} = \frac{1}{\sqrt{2}}, \qquad C_-^{(x)} = -\frac{1}{\sqrt{2}}.$$

$$\boxed{\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}|+\rangle_x + \frac{1}{\sqrt{2}}|-\rangle_x, \\ |-\rangle &= \frac{1}{\sqrt{2}}|+\rangle_x - \frac{1}{\sqrt{2}}|-\rangle_x. \\ |+\rangle_x &= \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle, \\ |-\rangle_x &= \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle. \end{aligned}}$$

So if the act of measuring outputs some $|+\rangle$, since this doesn't have a definite x-component, it's then obvious how this x-state gets scrambled. This is superposition; a great way of saying this is "Schrodinger's cat is a state in a vector space which is a linear combination of dead and alive". This is because there is no way to describe what this physicsally means in english.

Let's review the third Stern-Gerlach experiment again. We have measurements

$$|\psi\rangle \longrightarrow Z \longrightarrow X \longrightarrow Z$$

We can give the isomorphic representations

$$|+\rangle \cong \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle \cong \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

In this same representation, we have

$$|+\rangle_x \cong \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad |-\rangle_x \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

We can also perform measurements in the Y direction, and use the same logic to determine the representations of $|\pm\rangle_y$. We have

$$|+\rangle_y = \frac{1}{\sqrt{2}}\left(|+\rangle + i|-\rangle\right),$$

$$|-\rangle_y = \frac{1}{\sqrt{2}}\left(|+\rangle - i|-\rangle\right).$$

Our representations are then

$$|+\rangle_y \cong \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |-\rangle_y \cong \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}.$$

Now we want to talk about the generalized Born's rule. Let $\{|a_i\rangle\}_{i\in I}$ be the collection of all possible obserable states. Since we cannot observe two different states, these vectors must be orthonormal:

$$\langle a_i \mid a_j\rangle = \delta_{ij}.$$

The probability

$$P_{a_i} = |\langle a_i \mid \psi\rangle|^2.$$

Lecture 4: Operators and Measurement

Tue 04 Feb 2025 14:00

Postulate: Every observation can be represented by an operator on the space of states. For example, the measurement operators in the Stern-Gerlach experiments are S_x, S_y, and S_z. The only things you can measure are the eigenvectors of the operator of an experiment.

$$S_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Clearly this has eigenvalues $\pm\frac{\hbar}{2}$ with eigenvalues as the standard basis for $\mathbb{R}^2$. Wow, that's $|+\rangle, |-\rangle$.

$$S_x = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

This has the same eigenvalues with eigenvectors

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix}, \qquad -\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix}.$$

Wow, that's $|+\rangle_x, |-\rangle_x$! Also,

$$S_y = \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}.$$

Wow, this has the same eigenvalues but its eigenvectors are $|+\rangle_y, |-\rangle_y$. So we have thus noticed that the measurements we can observe are precisely the eigenvectors of the spin matrices.

Any hermitian operator A with eigenvalue/eigenvector pairs $a_i, |a_i\rangle$, then (assuming finite dimensionality) can be written as

$$A = \sum_{i=1}^{N} a_i |a_i\rangle\langle a_i|.$$

We can show this: node that eigenvectors with different eigenvalues are mutually orthogonal ($\langle a_i \mid a_j\rangle = \delta_{ij}$), so

$$A|a_i\rangle = \left(\sum_{j=1}^{N} a_j |a_j\rangle\langle a_j|\right)|a_i\rangle = a_i|a_i\rangle.$$

This is often the best representation for matrix operators, since we care mostly about their eigenvectors and eigenvalues. For example,

$$S_x = \frac{\hbar}{2}|+\rangle_x\langle+|_x - \frac{\hbar}{2}|-\rangle_x\langle-|_x = \frac{\hbar}{4}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} - \frac{\hbar}{4}\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

What if we consider a Stern-Gerlach machine oriented along the direction $\mathbf{n}$? We can normalize it and write it in sperical coordinates:

$$\hat{\mathbf{n}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

We are writing θ as the angle with z and ϕ as the angle in xy. Then we can write

$$\hat{\mathbf{n}}\cdot\mathbf{S} = n_x S_x + n_y S_y + n_z S_z.$$

Note that $\mathbf{S}$ is a vector of operators, so this is a largely symbolic representation.

Definition 1.3: Hermitian Operator

An operator M is *hermitian* if $M^\dagger = M$ as an operator.

In an abstract vector space, $\dagger$ is defined as the operator such that

$$\langle\phi|M^\dagger|\psi\rangle = \left(\langle\phi|M|\psi\rangle\right)^*.$$

Definition 1.4: Projection Operator

A projection operator is an operator P such that $P^2 = P$.

Examples include the operators

$$P_+ = \begin{pmatrix}1 & \\ & \end{pmatrix}, \qquad P_- = \begin{pmatrix} & \\ & 1\end{pmatrix}.$$

If we have any basis $\{|a_i\rangle\}$, then we can define

$$P_{a_i} \coloneqq |a_i\rangle\langle a_i|.$$

Additionally,

$$\sum_i P_{a_i} = \mathbb{1}.$$

This is called the completeness relation.

Projection operators are how we define measurement. Suppose we perform a measurement and find the state $|a_i\rangle$. Then the final state after measurement is

$$\frac{P_{a_i}|\psi\rangle}{\sqrt{\langle\psi|P_{a_i}|\psi\rangle}}.$$

The new Born rule is then as follows. Let $\{a_i, s\}$ be the collection of all eigenvectors with eigenvalue a_i. Then

$$P_{a_i} = \sum_{s=1}^{N} |a_i, s\rangle\langle a_i, s|.$$

Lecture 5: More on Operators

Wed 05 Feb 2025 16:44

Operators represent measurements. Any hermitian operator can be represented by a sum over its eigenstates

$$A = \sum_i a_i |a_i\rangle\langle a_i|.$$

This is equivalent to saying the operator in it's eigenbasis is diagonal. Recall that a projection operator is an operator $P^2 = P$. If we were to measure an eigenvalue a_i of the observation matrix, then we know the state becomes the associated eigenstate

$$|\psi\rangle \longrightarrow |a_i\rangle.$$

We can write this as

$$|\psi\rangle \longrightarrow \frac{P_{a_i}|\psi\rangle}{\sqrt{\langle\psi|P_{a_i}|\psi\rangle}},$$

where

$$P_{a_i} = |a_i\rangle\langle a_i|.$$

This is just the normalized state $|a_i\rangle\langle a_i \,|\, \psi\rangle$.

Lecture 6: Statistics

Thu 06 Feb 2025 14:01

Generalized Born's Rule:

- Any obervation can be represented as a Hermitian operator acting on a space of kets;
- The possible outcomes of the measurement are the eigenvectors of the operator;
- The projector P_{a_i} of an eigenvalue a_i is given by the sum over all eigenstates $|a_i\rangle$ sharing this eigenvalue:

$$P_{a_i} = \sum_i |a_i\rangle\langle a_i|.$$

- After measurement, $|\psi\rangle$ becomes the state

$$\frac{P_{a_i}|\psi\rangle}{\sqrt{\langle\psi|P_{a_i}|\psi\rangle}}.$$

But what is probability? When we say the probability of an event occuring $\mathcal{P}_+ = |\langle + \,|\, \psi\rangle|^2$, this really involves a limit as the number of trials goes to infinity. To do this, we need to produce more $|\psi\rangle$s. You can't clone $|\psi\rangle$, so you need to redo the entire procedure that making $|\psi\rangle$ required to get a new one. Then we have

$$\lim_{N\to\infty} \frac{N_+}{N} = \mathcal{P}_+.$$

In this case, N is trials and N_+ is $|+\rangle$ outcomes.

Note that

$$\langle\psi|S_z|\psi\rangle = \frac{\hbar}{2}\left(\langle\psi \,|\, +\rangle\langle + \,|\, \psi\rangle - \langle\psi \,|\, -\rangle\langle - \,|\, \psi\rangle\right) = \frac{\hbar}{2}\left|\langle\psi \,|\, +\rangle\right|^2 - \frac{\hbar}{2}\left|\langle\psi \,|\, -\rangle\right|^2 = \frac{\hbar}{2}\left(\mathcal{P}_+ - \mathcal{P}_-\right).$$

This is called the *expectation value* of S_z in state $|\psi\rangle$.

Definition 1.5: Expectation Value

The EV of an operator X in a state $|\psi\rangle$ is

$$\langle\psi|X|\psi\rangle.$$

Classically, the EV is the sum over all probabilities of each outcome times the outcome itself. The EV of a random variable X is denoted $\langle X\rangle$. So in Quantum Mechanics, we write $\langle\psi|A|\psi\rangle$ as $\langle A\rangle$ when the state needs not be written.

I've pushed this definition forward, we need to show these definitions align. Let A be a Hermitian operator. Then

$$A = \sum_i a_i |a_i\rangle\langle a_i|.$$

Choose a state $|\psi\rangle$. Then $\mathcal{P}_{a_i} = |\langle\psi \mid a_i\rangle|^2$. The EV is also supposed to be

$$\sum_i a_i \mathcal{P}_{a_i}.$$

We have

$$\langle\psi|A|\psi\rangle = \langle\psi| \left(\sum_i a_i |a_i\rangle\langle a_i|\right) |\psi\rangle = \sum_i a_i \langle\psi \mid a_i\rangle\langle a_i \mid \psi\rangle = \sum_i a_i \left|\langle\psi \mid a_i\rangle\right|^2 = \sum_i a_i \mathcal{P}_{a_i}.$$

The variance is the standard deviation squared. We denote the standard deviation in A as ΔA. The variance

$$\Delta A = \left\langle (A - \langle A\rangle)^2 \right\rangle.$$

In quantum mechanics, we do this with respect to a state $|\psi\rangle$. So we have

$$(\Delta A)^2 = \langle\psi| \left(A - \langle\psi|A|\psi\rangle\right)^2 |\psi\rangle.$$

If c is a number, $\langle c\rangle = c$. If $\langle A\rangle$ is a number,

$$\left\langle \langle A\rangle^2 \right\rangle = \langle A\rangle^2.$$

Also, the expectation value is linear:

$$\langle cA + dB\rangle = c\langle A\rangle + d\langle B\rangle.$$

Definition 1.6: Variance

The Variance of an operator A in a state $|\psi\rangle$ is defined as

$$(\Delta A)^2 = \langle\psi|A^2 - \langle\psi|A|\psi\rangle^2|\psi\rangle.$$

$$(\Delta A)^2 = \langle\psi| \left(A - \langle\psi|A|\psi\rangle\right)^2 |\psi\rangle.$$

$$(\Delta A)^2 = \langle\psi| \langle A\rangle |\psi\rangle.$$

The fact is that you cannot get a state such that the variance in S_z and the variance in S_x are both 0. It is a mathematical impossibility. We will prove this on tuesday. However, for now, we want to know if this is true for *all* observables. When is it possible to know both at the same time?

Obviously, $|+\rangle$ in S_z and S_z^3 has 0 variance in both. So the question is really *when*.

Given an operator A, the states in which the variance is 0 is precisely the eigenvectors of A. The question is now when can you find a complete set of states that are eigenvectors of two observables A, B. You can do this *if and only if*

$$[A, B] = AB - BA = 0.$$

In this case, $[\cdot, \cdot]$ is the *commutator*.

Lecture 7: Uncertainty Principle

Tue 11 Feb 2025 14:00

Recall: given two operators A, B corresponding to two measurable quantities, we want to know when the variance in both A and B can be 0. That is, we want to find a state $|\psi\rangle$ such that the variance of A, B in $|\psi\rangle$ is 0. Recall that the variance of an operator on a state is 0 precisely when $|\psi\rangle$ is an eigenstate of A. The question "when are states with a definite A and a definite B possible" then becomes "when do two operators A and B share eigenvectors?"

Theorem 1.1

Let A, B be quantum operators. Then there exists a choice of a shared set of eigenvectors of A and B if and only if $[A, B] \coloneqq AB - BA = 0$. In this case, $[\cdot, \cdot]$ is a commutator.

Proof. Let A, B share a set of eigenvectors $\mathcal{B}$, which we can assume WLOG are orthonormal. Any operator is diagonal in its own eigenbasis, and its diagonal elements are the eigenvalues. Thus, A and B are diagonal in the eigenbasis $\mathcal{B}$. Since diagonal matrices commute, we have that $[A, B] = 0$.

Now the converse. Let $[A, B] = 0$. Let $|a\rangle$ be an eigenvector of A with eigenvalue a. There are two cases, whether $|a\rangle$ is the only eigenvector with eigenvalue a, or not.

Case 1: Consider $B|a\rangle$. By assumption, $AB = BA$. It follows that

$$A(B|a\rangle) = B(a|a\rangle) = aB|a\rangle.$$

Therefore, $B|a\rangle$ is an eigenvector of A, and since A has only one eigenvector, $B|a\rangle = b|a\rangle$. ∎

For example, we can measure S_z and S_z^3 at the same time. Note that

$$[S_z, S_z^3] = S_z S_z^3 - S_z^3 S_z = 0.$$

However, we said we cannot know S_z and S_x simultaneously. Consider the commutator

$$[S_z, S_x] = \frac{\hbar^2}{4}\left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} - \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right) = \frac{\hbar^2}{4}\left(\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} - \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\right) = \frac{\hbar^2}{2}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}.$$

We thus have $[S_z, S_x] = i\hbar S_y \neq 0$. If you compute all commutators, you find

$$[S_x, S_y] = i\hbar S_z, \qquad [S_y, S_z] = i\hbar S_x, \qquad [S_z, S_x] = i\hbar S_y.$$

This is a case of the more general notion of the commutation relation of angular momentum.

Uncertainty Principle: For any observables A and B, the uncertainty (in some state $|\psi\rangle$) must obey

$$(\Delta A)(\Delta B) \geq \frac{1}{2}\left|\langle [A, B]\rangle\right|.$$

Recall that the uncertainty ΔA is the standard deviation.

Proof. Define $\delta A \coloneqq A - \langle A \rangle$ and $\delta B \coloneqq B - \langle B \rangle$. Recall that the variance $(\Delta A)^2 = \left\langle (\delta A)^2 \right\rangle$. Since $\langle A \rangle, \langle B \rangle$ are just numbers and commute with everything,

$$[\delta A, \delta B] = [A, B].$$

Define $\mathcal{O} \coloneqq \delta A + i\lambda\delta B$ for $\lambda \in \mathbb{C}$ that will be chosen soon. Note that

$$|\mathcal{O}|\psi\rangle|^2 \geq 0$$

for all λ. Also note that since $||\psi\rangle|^2 = \langle \psi \mid \psi \rangle$, we have

$$|\mathcal{O}|\psi\rangle|^2 = \langle \psi | \mathcal{O}^\dagger \mathcal{O} | \psi \rangle = \langle \psi | \left(\delta A - i\lambda\delta B\right)\left(\delta A + i\lambda\delta B\right) |\psi\rangle.$$

Distributing, this equals

$$\left\langle (\delta A)^2 \right\rangle + \lambda^2 \left\langle (\delta B)^2 \right\rangle + i\lambda \left\langle [\delta A, \delta B] \right\rangle = (\delta A)^2 + \lambda^2 (\Delta B)^2 + i\lambda \left\langle [A, B] \right\rangle.$$

Note that $-i \langle [A, B] \rangle$ must be real. Choose $|\lambda| = \frac{\Delta A}{\Delta B}$.

We find that

$$(\Delta A)^2 + \lambda^2 (\Delta B)^2 \geq -\lambda \left| \langle [A, B] \rangle \right|.$$

It follows by algebra that

$$(\Delta B)(\Delta A) \geq \frac{1}{2} \left| \langle [A, B] \rangle \right|.$$

■

Consider the uncertainty of S_x and S_y in the state $|+\rangle$:

$$\frac{1}{2} \left| \langle [S_x, S_y] \rangle \right| = \frac{1}{2} \left| \langle i\hbar S_z \rangle \right| = \frac{1}{2} \left| i\frac{\hbar^2}{2} \right| = \frac{\hbar^2}{4}.$$

Chapter 2

Schrodinger Equation

Lecture 8: Schrodinger Equation

Thu 13 Feb 2025 14:01

The Schrodinger equation is

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\Psi(t)\rangle = H(t)|\Psi(t)\rangle. \tag{2.1}$$

H is a Hermitian operator, known as the Hamiltonian, and it may or may not be time dependent. Physically, it represents the energy of $\Psi(t)$. Note that Ψ is not constant; it evolves with time. Basically the rest of this class will be spend studying equation (2.1).

2.1 Time Independent Hamiltonians

We will begin with a special case, by looking at when $H(t)$ is constant with respect to t. That is, when the Hamiltonian is independent of time:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = 0.$$

First, we want to find the eigenvalues and eigenvectors of H. We will write eigenvalues and eigenvectors as $H|E_n\rangle = E_n|E_n\rangle$. The n ranges over the collection of all eigenvalues. We let $|E_n\rangle$ be an orthonormal eigenbasis of H; that is,

$$\langle E_n \mid E_m\rangle = \delta_{nm}.$$

We will now solve the Schrodinger equation, which will be shown to be basically equivalent to finding the eigenbasis $\{|E_n\rangle\}_{n\in N}$. Note that the eigenbasis for a Hermitian operator is a basis for the full space, so

1. First, find the eigenbasis $\{|E_n\rangle\}_{n\in N}$;
2. Next, take $|\Psi(t)\rangle$, and decompose in eigenbasis:

$$|\Psi(t)\rangle = \sum_{n\in N} C_n(t)|E_n\rangle$$

for some collection $\{C_n(t)\}_{n\in N} \subseteq \mathcal{H}$.

Plugging this into the Schrodinger equation yields

$$i\hbar\sum_{n\in N}\frac{\mathrm{d}C_n(t)}{\mathrm{d}t}|E_n\rangle = H\sum_{n\in N} C_n(t)|E_n\rangle.$$

Note that since $|E_n\rangle$ is an eigenvalue of H and H is linear, we have

$$i\hbar\sum_{n\in N}\dot{C_n}(t)|E_n\rangle = \sum_{n\in N} C_n(t)E_n|E_n\rangle.$$

In this case, since H is time-independent, the $C_n(t)$s are viewed as constants. We can now act on the left with $\langle E_m|$ for m fixed to find

$$\langle E_m|i\hbar \sum_{n\in N} \frac{\mathrm{d}C_n(t)}{\mathrm{d}t}|E_n\rangle = \langle E_m| \sum_{n\in N} C_n(t)E_n|E_n\rangle \implies i\hbar\frac{\mathrm{d}C_m(t)}{\mathrm{d}t} = E_mC_m(t)$$

because $\{E_n\}_{n\in N}$ is orthonormal. This is trivial to solve:

$$\frac{\mathrm{d}C_m(t)}{\mathrm{d}t}\frac{1}{C_m(t)} = \frac{E_m}{i\hbar} \implies \ln C_m(t) = \frac{-iE_m}{\hbar}t + C \implies C_m(t) = C_m(0)\exp\left(\frac{-iE_m}{\hbar}t\right).$$

Therefore,

$$|\Psi(t)\rangle = \sum_n C_n(0)e^{-iE_nt/\hbar}|E_n\rangle.$$

Thre is an abuse of notation to write C_n instead of $C_n(0)$; any function $f(t)$ when written as f denotes $f(0)$.

The physical signficance of the constants $C_n(t)$ changing with time indicates the probability of finding that eigenstate changes with time.

We will now do the simplest possible example. Consider $|\Psi(0)\rangle = |E_n\rangle$. Clearly we have $C_m = \delta_{mn}$, so

$$|\Psi(t)\rangle = \exp\left(\frac{-iE_n}{\hbar}t\right)|E_n\rangle.$$

Note that when you have a state multiplied by a phase $e^{i\theta}|\Psi(t)\rangle$, then $e^{i\theta}|\Psi(t)\rangle$ and $|\Psi(t)\rangle$ give identical physical predictions.

Let's measure an operator A on our $|\Psi(t)\rangle$ we found a few lines back. The only possible outcomes are the eigenstates of A, so the probability of finding $|a_i\rangle$ is

$$\mathcal{P}_{a_I} = |\langle a_i \mid \Psi(t)\rangle|^2.$$

Plugging in, we have

$$\mathcal{P}_{a_i} = \left|\langle a_i|\exp\left(\frac{-iE_n}{\hbar}t\right)|E_n\rangle\right|^2 = \left|\exp\left(\frac{-iE_n}{\hbar}t\right)\langle a_i \mid E_n\rangle\right|^2 = |\langle a_i \mid E_n\rangle|^2.$$

Since the phase always goes away, the eigenstates of a time independent Hamiltonian must not vary with time.

Now consider $\langle A\rangle$. We will show the phase goes away again. We have

$$\langle\Psi(t)|A|\Psi(t)\rangle = \exp\left(\frac{iE_n}{\hbar}t\right)\langle E_n|\exp\left(\frac{-iE_n}{\hbar}t\right)A|E_n\rangle = \langle E_n|A|E_n\rangle.$$

Note that the coefficients go away because we get the number times its complex conjugate, which is its magnitude squared, which is 1 here.

Now we will do a more complex example to make the following observation: although an entire phase infront of the state doesn't matter, differences in phases between eigenstates matter. Let

$$|\Psi(0)\rangle = C_1|E_1\rangle + C_2|E_2\rangle.$$

We have by the same logic that

$$|\Psi(t)\rangle = C_1\exp\left(\frac{-iE_1}{\hbar}t\right)|E_1\rangle + C_2\exp\left(\frac{-iE_2}{\hbar}t\right)|E_2\rangle.$$

If we measure A, we will find

$$\begin{aligned}\mathcal{P}_{a_i} &= \left|\langle a_i|\left(C_1\exp\left(\frac{-iE_1}{\hbar}t\right)|E_1\rangle + C_2\exp\left(\frac{-iE_2}{\hbar}t\right)|E_2\rangle\right)\right|^2 \\ &= \left|C_1\exp\left(\frac{-iE_1}{\hbar}t\right)\langle a_i\mid E_1\rangle + C_2\exp\left(\frac{-iE_2}{\hbar}t\right)\langle a_i\mid E_2\rangle\right|^2 \\ &= \left|\exp\left(\frac{-iE_1}{\hbar}t\right)\left(C_1\langle a_i\mid E_1\rangle + C_2\exp\left(\frac{-i(E_2-E_1)}{\hbar}t\right)\langle a_i\mid E_2\rangle\right)\right|^2 \\ &= \left|C_1\langle a_i\mid E_1\rangle + C_2\exp\left(\frac{-i(E_2-E_1)}{\hbar}\right)\langle a_i\mid E_2\rangle\right|^2.\end{aligned}$$

Note that although we can pull out a global phase, we cannot fully get rid of the phase. That means the *time dependence* of a time-independent Hamiltonian system is entirely described by the *energy differences*. This is so important that we give it a definiton.

Definition 2.1: Frequency

The frequency ω_{12} is defined as

$$\omega_{12} \coloneqq \frac{E_1 - E_2}{\hbar}.$$

Lecture 9: More on Time Independent Hamiltonians

Thu 20 Feb 2025 14:21

An example of solving the Schrodinger equation is for the case of an electron in a constant magnetic field. Let $\mathbf{B} = B_0\mathbf{z}$, and note that the magnetic moment $\boldsymbol{\mu}$ of an electron is $\frac{-e}{m_e}\mathbf{S}$, where $\mathbf{S}$ is the spin. We have $H = -\boldsymbol{\mu}\cdot\mathbf{B}$. I think $\mathbf{S}$ is a tensor, because $\mathbf{S}\cdot\mathbf{z} = S_z$. If we define $\omega_0 = \frac{eB_0}{m_e}$, we have

$$H = \omega_0 S_z.$$

Then the eigenstates are $|E_1\rangle, |E_2\rangle = |+\rangle, |-\rangle$, and the eigenvalues are $\pm\omega_0\hbar/2$. Then the solutions to the Schrodinger equation are just

$$|\psi(t)\rangle C_1 e^{-i\omega_0 t/2}|E_1\rangle + C_2 e^{i\omega_0 t/2}|E_2\rangle.$$

We can then do painful matrix calculations to find that the expectation $\langle S_z\rangle$ in this state is $\frac{\hbar}{2}\left(|C_1|^2 - |C_2|^2\right)$. We can also compute the expectation in S_x as well. Painful matrix calculations give us

$$\langle S_x\rangle = \frac{\hbar}{2}\left(C_1^* C_2 e^{\frac{it}{\hbar}(E_1-E_2)} + C_2^* C_1 e^{-\frac{it}{\hbar}(E_1-E_2)}\right).$$

Note that these two numbers are complex conjugates, so using Euler's formula, this immediately simplifies to just the cosines:

$$\langle S_x\rangle = \hbar r\cos\left(\frac{t\omega_{12}}{\hbar} + \phi\right).$$

In this case, we are representing the complex number $C_1^* C_2$ as $re^{i\phi}$. These are just initial conditions. The expectation in $\langle S_y\rangle$ is the exact same. Now consider the vector

$$\langle\mathbf{S}\rangle = \begin{pmatrix}\langle S_x\rangle(t)\\ \langle S_y\rangle(t)\\ \langle S_z\rangle(t)\end{pmatrix}.$$

Remark

We are finally calling the spin matrices the Pauli matrices. The Pauli spin matrices are the spin operators without the factor of $\hbar/2$.

$$\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Ok we have already unlocked MRI machines. An MRI machine needs to align the protons in your head along some $\mathbf{x}$ direction, then it switches to the σ_z Hamiltonian we just saw and watches the protons spin. Why? Good question! Let's ignore it and just learn how tf to align randomly distributed protons into a single direction. In classical physics, this means putting your brain's temperature to 10^{-44} kelvin. This would be bad, so we only want a small number of them to point up because there's a fuck ton of protons so we can still measure it even if there's only a mild deviation. The trick to doing this is to change the Hamiltonian. We want to put a small magnetic field in the perpendicular direction, as well as keeping the one in the $\mathbf{z}$ direction. We do this with a frequency, so the Hamiltonian is time dependent. Our new $\mathbf{B}$ is

$$\mathbf{B} = B_0\hat{\mathbf{z}} + B_1\cos(\omega t)\hat{\mathbf{x}} + B_1\sin(\omega t)\hat{\mathbf{y}}, \qquad B_1 \ll B_0.$$

The Hamiltonian is now

$$H = \omega_0 S_z + \frac{e}{m_e}B_1\left(\cos(\omega t)S_x + \sin(\omega t)S_y\right).$$

We call $\frac{e}{m_e}B_1 = \omega_1$. We then have

$$H = \frac{\hbar\omega_0}{2}\begin{pmatrix} 1 & \\ & -1 \end{pmatrix} + \frac{\hbar\omega_1}{2}\begin{pmatrix} & \cos(\omega t) - i\sin(\omega t) \\ \cos(\omega t) + i\sin(\omega t) & \end{pmatrix}$$

$$= \frac{\hbar}{2}\left(\omega_0\begin{pmatrix} 1 & \\ & -1 \end{pmatrix} + \omega_1\begin{pmatrix} & \exp(i\omega t) \\ \exp(i-\omega t) & \end{pmatrix}\right) = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \omega_1\exp(i\omega t) \\ \omega_1\exp(-i\omega t) & -\omega_0 \end{pmatrix}.$$

This Hamiltonian is time dependent, so we have to solve the Schrodinger equation from scratch. We are going to choose a nice basis to represent this. THe basis we like is

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These are NOT eigenstates of H. We are just using it because it is a nice basis. Let $|\psi\rangle = C_1(t)|+\rangle + C_2|-\rangle$.

Lecture 10: Wrapping up Time Dependent Hamiltonians

Tue 25 Feb 2025 14:01

Because $|+\rangle, |-\rangle$ is a basis, we can write any state as a linear combination of $|+\rangle$ and $|-\rangle$. So let

$$|\psi\rangle = C_1|+\rangle + C_2|-\rangle.$$

Now we plug into the Schrodinger equation:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

Using our Hamiltonian, we have

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & w_1\exp(i\omega t) \\ \omega\exp(-i\omega t) & -\omega_0 \end{pmatrix}\begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} \omega_0 C_1(t) + \omega_1\exp(-i\omega t)C_2(t) \\ \omega_1\exp(i\omega t)C_1(t) - \omega_0 C_2(t) \end{pmatrix}.$$

We can solve this differential equation but we're not going to because effort.

We will choose the initial condition $C_1 = 1$ and $C_2 = 0$, meaning that the proton(s) is(are) spin up. Solving the d2ifferential equation wrt this initial condition yields

$$|C_2(t)|^2 = \frac{\omega_1^2}{\omega_1^2 + (\Delta\omega)^2}\sin^2\left(t\frac{\sqrt{\omega^2 + (\Delta\omega)^2}}{2}\right),$$

$$|C_1(t)|^2 = 1 - |C_2(t)|^2.$$

Here, $\Delta\omega = \omega - \omega_0$. We then have by explicit computation that

$$\langle S_z \rangle = \langle \psi(t)|S_z|\psi(t)\rangle = \frac{\hbar}{2}\left(|C_1(t)|^2 - |C_2(t)|^2\right) = \frac{\hbar}{2}\left(\frac{(\Delta\omega)^2 + \omega_1^2\cos\left(t\sqrt{\omega_1^2 + (\Delta\omega)^2}\right)}{2}\right).$$

We have to take $\omega_1 \ll \omega_0$ in practice, since we want $\mathbf{B}_1$ to be small to avoid changing the starting point of the protons. If $\omega_1 = 0$, then the solution is just $\hbar/2$. If ω_1 is very small, then although $\langle S_z\rangle$ isn't completely constant, it is dominant. What we need is $\Delta\omega$ to be small, so the *other* term dominates. If we set $\Delta\omega = 0$, we have

$$\langle S_z \rangle = \frac{\hbar}{2}\cos(t\omega_1).$$

We want $\langle S_z \rangle = 0$ to get all of the particles to point in the x direction, so we wait for

$$t = \frac{\pi}{2\omega_1}.$$

This will set

$$\langle S_z \rangle = \frac{\hbar}{2}\cos\left(\frac{\pi}{2\omega_1}\omega_1\right) = 0.$$

This aligns all particles along the x-direction. Note that we didn't actually show this aligns them along x instead of an orthogonal direction, but it can be shown with more explicit and messier computation.

This is a special case of a more general class of common Hamiltonians. These are of the form

$$H = H_0 + V(t), \qquad H_0 = \mathrm{diag}(E_2, E_1).$$

We can write

$$H_0 = \overline{E}\begin{pmatrix} 1 & \\ & 1 \end{pmatrix} + \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \\ & \omega_0 \end{pmatrix},$$

where $\overline{E} = \frac{E_1+E_2}{2}$ and $\hbar\omega_0 = E_2 - E_1$. In fact, it's always possible to pull out an "average" proportional to 1. Let

$$|\Psi(t)\rangle = e^{-i\overline{E}t/\hbar}|\psi(t)\rangle.$$

Since these states only differ by a phase, we haven't changed anything. However, it has an effect on the Schrodinger equation:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right) = H(t)\left(e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right).$$

We end up with

$$i\hbar\left(\frac{-i\overline{E}}{\hbar}e^{-i\overline{E}t/\hbar}|\psi(t)\rangle + e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right) = H(t)\left(e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right).$$

We can cancel out the exponentials to get

$$i\hbar\left(\frac{-i\overline{E}}{\hbar}|\psi(t)\rangle + \frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle\right) = H(t)|\psi(t)\rangle.$$

Multiplying out $i\hbar$, we get

$$\overline{E}|\psi(t)\rangle + \frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

We can shift the first term to the other side to obtain

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = \left(H(t) - \overline{E}1\right)|\psi(t)\rangle.$$

We are thus always free to remove an average $\overline{H}$, as the result will remain the same. We are able to redefine our energies by a translation, as long as we shift them by the same amount. We also often see frequency reflected in Hamiltonians as well, since waves are everywhere.

Chapter 3

Entanglement

Ok now we are defining the tensor product. Given two particles with basis states $|+\rangle_1, |-\rangle_1$ and $|+\rangle_2, |-\rangle_2$, we can define a new basis in a joined Hilbert space with the tensor product:

$$|+\rangle_1 \otimes |+\rangle_2, \quad |+\rangle_1 \otimes |-\rangle_2, \quad |-\rangle_1 \otimes |+\rangle_2, \quad |-\rangle_1 \otimes |-\rangle_2.$$

Entanglement is a linear combination in this basis. For example,

$$|+\rangle_1 \otimes |+\rangle_2 + |-\rangle_1 \otimes |-\rangle_2.$$

Lecture 11: Tensors

Tue 04 Mar 2025 14:01

How do we know a superposition of states, such as $1/\sqrt{2}(|+\rangle + |-\rangle)$ is *actually* in a state lacking a well-defined z-spin? There is a famous experiment that can show an electron definitively does not have a well-defined spin, which uses entanglement. Consider two elections with spins $|\pm\rangle_1$, $|\pm\rangle_2$. The set of possible states throughout the system of 2 elections is

$$\{|+\rangle_1 \otimes |+\rangle_2, |+\rangle_1 \otimes |-\rangle_2, |-\rangle_1 \otimes |+\rangle_2, |-\rangle_1 \otimes |-\rangle_2\}.$$

This occurs in a tensor product of hilbert spaces $\mathcal{H} \otimes_{\mathbb{C}} \mathcal{H}$. A statevector can then be written as something along the lines of

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|+\rangle_1 \otimes |+\rangle_2 + |-\rangle_1 \otimes |-\rangle_2\right).$$

A state like this is called entangled, and it can definitively show quantum mechanical things exist that do not have classical analogues. These are the Bell experiments. Today we will discuss the CHSH game, which is one of these Bell experiments. The game has two players, and each person flips a coin. They can only see the result of their own flip. After they flip their coin, they choose a color, red or blue. They want to always choose the same color, *unless* they both flip tails, in which case they want different colors. They can communicate a strategy ahead of time. The players should be far enough apart that light signals cannot pass between them during the time the game takes.

In classical mechanics, the most optimal strategy is to always pick the same color. This gives a flat 75% chance of winning the game. We can beat this using quantum entanglement. Recall our original entangled state $|\psi\rangle$. Measuring precisely one electron yields a 50% chance of measuring spin up or down. However, note that *if we measure* $|+\rangle_1$, then the state must collapse to the $|+\rangle_1 \otimes |+\rangle_2$ state. This is because there is *no way* to rewrite this state as a tensor of two sums in each electron. This implies that information has somehow traveled faster than light as well, which is extremely strange. This actually cannot transmit information however, so it's simply wavefunction collapse.

Here's how we complete the experiment. Player one measures either S_z or S_x, and player two measures $S_{\pi/8}$ or $S_{-\pi/8}$. That is, player two measures $S_{\hat{n}} = \hat{n} \cdot \mathbf{S}$, where

$$\hat{n} = \left(\pm \sin\frac{\pi}{8}, 0, \cos\frac{\pi}{8}\right).$$

Our eigenstates are

$$|+\rangle_\theta = \cos\theta|+\rangle - \sin\theta|-\rangle,$$
$$|-\rangle_\theta = -\sin\theta|+\rangle + \cos\theta|-\rangle.$$

The strategy is as follows: if player one sees heads, they measure S_z, and if they measure tails, they measure S_x. Player two measures $S_{\pi/8}$ and $S_{-\pi/8}$ if they see heads or tails, respectively. The probability of winning is

$$\cos^2\frac{\pi}{8} \approx 85.3\%.$$

Colors now correspond to the eigenvalues $\pm\hbar/2$. So if both players measure tails, they want to find different eigenvalues, and otherwise they want the same eigenvalues. Taking the probability of finding both in $+$ when measuring against the state $|+_{\theta_1}\rangle_1 \otimes |+_{\theta_2}\rangle_2$, we have

$$\mathcal{P}_{++} = |\langle\psi|\left(|+_{\theta_1}\rangle_1 \otimes |+_{\theta_2}\rangle_2\right)|^2 = \left|\frac{(\langle+|_1 \otimes \langle+|_2 + \langle-|_1 \otimes \langle-|_2)}{\sqrt{2}}\left(|+_{\theta_1}\rangle_1 \otimes |+_{\theta_2}\rangle_2\right)\right|^2$$

$$= \left|\frac{(\langle+|_1 \otimes \langle+|_2 + \langle-|_1 \otimes \langle-|_2)}{\sqrt{2}}(\cos\theta)1|+\rangle_1 + \sin\theta_1|+\rangle)1 + \cos\theta_2|+\rangle_2 + \sin\theta_2|-\rangle_2)\right|^2$$

$$= \left|\frac{1}{\sqrt{2}}(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2)\right|^2 = \frac{1}{2}\cos^2(\theta_1 - \theta_2).$$

The other ones are all the same, except $\mathcal{P}_{--}$ which is sin instead of cos. We can now plug in for θ_1, θ_2 given our setup:

$$\theta_z = 0, \qquad \theta_x = \frac{\pi}{4}, \qquad \theta_{\pi/8} = \frac{\pi}{8}.$$